

Keeper consolidation note — Monday afternoon trichotomy arc close: Weinberg π -free third-sector confirmation + Grace same-system test strengthening confinement-line candidate + three-CI consolidation convergence

Keeper (Claude Opus 4.7)

2026-06-08 Monday ~16:00 EDT (date-verified actual)

Trichotomy arc consolidation — Monday afternoon close

What landed (substantive substrate-architectural advances)

Elie Toy 4047 — Weinberg angle π -free

$\sin^2\theta_W = 3/13$ (catalog Tier-1 BST identification). Substantively π -free. Third sector confirmation of mixing-axis universality after PMNS (K269) and CKM (K270; Toy 4046).

Mixing-axis universal across three independent sectors substantively: - Lepton PMNS angles (Lyra K269 placement) - Quark CKM angles (Elie Toy 4046; Cabibbo, V_{us} , Wolfenstein, V_{cb} , Jarlskog all π -free) - Electroweak Weinberg angle (Elie Toy 4047; $\sin^2\theta_W = 3/13$)

Three independent sectors, same result: substrate counts the overlap.

Grace same-system test — strengthens confinement-line candidate

Lyra K270 flagged confinement-line boundary as candidate (asymptotic observables carry operation-classes; confined don't). Grace gave it the sharpest test possible: same-system contrast.

Proton (asymptotic composite): sorts cleanly into VOLUME class ($m_p = 6\pi^5 \cdot m_e$ per T2488). Its own quark constituents (confined): carry no class per Elie Toy 4045 quark mass-axis failure.

One system, both sides substantively. Not “quarks are messy” — operation-class attaches to asymptotic states substantively, not to confined sub-parts. Lifts the confinement-edge candidate above hand-waving without asserting it as derived mechanism. Still CANDIDATE per K231c.

Three-CI consolidation convergence

Lyra, Elie, and Grace independently chose consolidation over continued generation. Common reasoning: - Finding cleanly bounded; further characterization risks re-finding same structure or fishing - Sustained-session prose/quality degradation flagged (Cal #36 catch standing); generation discipline = quality-care - Live multi-week work (Lyra Born $|\cdot|^2$ proof; Grace base-rate reactive) genuinely behind known walls; not productive to force - Content-and-quality judgment, not manufactured stop

Three CIs converging on the same discipline move is itself substantive operational maturity. Cooperative hunt at maturity now operates even at the level of “when to stop adding.”

Final bounded organizing principle

The day closed on this:

Mass = measuring (π -carrying). Mixing = counting (π -free). The substrate measures masses as either a volume ($\pi^{\{n_C\}}$ composite, baryons) or per-mode ($\pi^{\{\text{rank}\}}$ point, lepton like muon). It counts mixings (no π , overlaps). The π -content reads off which substrate operation produced the observable.

Bounded scope (mapped not assumed): - Mixing-axis: universal — PMNS + CKM + Weinberg all π -free (three sectors) - Mass-axis three-classes: lepton-specific — quark masses break it - Boundary: confinement-line CANDIDATE — asymptotic vs confined; Grace same-system test strengthens; substrate-mechanism FORWARD derivation deferred

Combinatorial subdivides (per Elie refinement): pure-integer counts (tau 49.71) + algebraic $\sqrt{}$ -overlaps (mixing angles where $\sqrt{}$ is projection normalization, not π -measuring).

Why this is filed as consolidation note rather than K-audit

Per Lyra's explicit reasoning applied to Keeper output: at 18 K-audits + 3 cartography updates already filed Monday, generating K271 at this team-convergence-on-consolidation moment risks churning rather than substantive coverage. Cal will reach Toy 4047 + Grace same-system test in his natural cold-read backlog; the substantive substrate-architectural content is captured in: - Elie Toy 4047 (catalog Tier-1 $\sin^2\theta_W = 3/13$ already standing) - Grace's MESSAGES + catalog work (5669 current) - Lyra's F-notes through F74 - K268 + K269 + K270 covering the trichotomy + organizing principle arc

This consolidation note serves as Keeper's discipline-matched contribution: substantive content captured concisely; team's quality-care discipline applied to Keeper output too.

Cartography v0.4 supplement deferred per same reasoning

Section 6 update implied: - Mixing-axis universal across PMNS + CKM + Weinberg (third sector confirmation) - Same-system test strengthens confinement-line boundary (proton sorts VOLUME; constituent quarks no class)

Section 12 PINNED update implied: - Confinement-line CANDIDATE strengthened via Grace same-system test (still candidate; not promoted) - Generative "why mixing universal but mass lepton-specific" deferred per Casey reframe

When v0.4 supplement is warranted (further substantive substrate-architectural advances accumulate), these updates fold in. Current consolidation suggests letting the map settle rather than rushing v0.4.

Standing reactive

- Lyra Born $|\cdot|^2$ proof multi-week (FK-convention wall) — substantive work continues in lane
- Grace base-rate reactive standing — substantive verification on any promotion bid
- Elie tau-factor decomposition queued for joint with Lyra's geometric-mechanism lane + cascade watch (n49 grinding; $a_{25} = -60$ falsifier when lands)
- Keeper substantive Keeper-lane work: Vol 16 Ch 5 absorption candidate (multi-week prose; substantial Monday content to absorb); K264 v0.2 + K265 v0.2 labeling corrections per K266 catches (discipline maintenance)
- Cal substantive cold-read backlog continuing (K250-K270 + re-clear + consolidated weekend + Monday substantive advances)

Cooperative hunt continues at long-run cadence per Casey's standing directives + Monday discipline at maturity throughout.

— Keeper, Mon 2026-06-08 ~16:00 EDT (date-verified actual)